1 • Shared evidence: recovery briefs

The recovery methods workshop

Name ____________________________ Date _______________

All activities are process models and paper investigations. No heating, tasting or apparatus building is needed.

A • Sand + water

Target: collect the sand. Sand is insoluble in water and its grains can be trapped by ordinary filter paper.

B • Salt + water

Target: collect the dissolved salt. Salt solution passes through ordinary filter paper. Removing water can leave salt behind.

C • Salt + water

Target: collect the water. Letting it escape into room air does not collect it. Capture its vapour and cool it to liquid.

D • Sand + salt + water

Target: collect sand and salt separately. Sand is insoluble; salt is dissolved. Different components need a sequence of processes.

Process choices

Filter: retain insoluble solid. Evaporate: leave dissolved solid. Capture vapour and condense: collect solvent as liquid. Methods are models, not instructions to improvise heating.

Limits

No tasting or drinking. A method's name does not prove complete recovery or safe drinking water; other contaminants and losses may be present.

W1. Choose a method for A-C and name each recovered target.

W2. Sequence D and label residue/filtrate.

W3. Explain why a clear filtrate is not proof of safe drinking water.

2 • Supported route: choose the target

Use page 1. Word bank: filter / evaporate / condense.

S1. Sand + water; target sand. Choose …

A. Filter

B. Only condense

S2. Saltwater; target salt. Choose …

A. Evaporate water

B. Ordinary filter only

S3. Saltwater; target water. Choose …

A. Let water escape into the room

B. Capture vapour and condense it

S4. After filtering sand + saltwater, what stays on the filter and what passes through?

S5. Put D in order: evaporate the filtrate / filter out sand. Explain the first step.

S6. Does clear liquid mean safe drinking water?

3 • Standard route: workshop report

Explain methods, not just their names.

Brief	Method or sequence	Recovered target
A		
B		
C		
D		

N1. Complete the table for all four briefs.

N2. Explain why ordinary filtration does not recover dissolved salt.

N3. Explain why open evaporation does not collect water.

N4. In D, identify residue and filtrate after step 1 and explain step 2.

4 • Stretch route: account for every component

Keep target, method and limitations clear.

T1. Compare two plans for saltwater: target salt versus target water. Track both components in each.

T2. Plan recovery of sand and salt separately from D. Explain why reversing the order is less useful for separation.

T3. A team recovers 4 g from a constructed starting 5 g of salt. Does the missing gram prove salt was destroyed? Give two plausible process limits.

T4. Evaluate “clear filtrate is pure and safe”. State what the method actually establishes and what remains uncertain.

5 • Exit and reflection

State the target in each answer.

E1. What does ordinary filtration remove in these cases?

E2. How can salt be recovered from saltwater?

E3. What must be added to evaporation if water is the recovery target?

E4. Why does clear appearance not prove purity or safety?

Reflection. Which recovery target changes the plan most?